

KEYU HE

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Education

Carnegie Mellon University, Pittsburgh, PA

Master of Science in Intelligent Information Systems @ School of Computer Science

May 2027

GPA: 4.14/4.33

University of Southern California, Los Angeles, CA

Bachelor of Science in Computer Science & Bachelor of Art in Applied Mathematics

May 2025

GPA: 3.98/4.00

Minor in Artificial Intelligence Applications

Skills

Programming: C++, C, Python, Java, MySQL, HTML, CSS, JS, x86-64 Assembly

Frameworks/Tools/Software: PyTorch, Pandas, NumPy, LangChain, FAISS, Git, AWS, L^AT_EX

Areas of Expertise: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Vision Language Models (VLMs), Data Science / Data Engineering

Publications

1. **Keyu He**, Tejas Srinivasan, Brihi Joshi, Xiang Ren, Jesse Thomason, Swabha Swayamdipta. *Believing without Seeing: Quality Scores for Contextualizing Vision-Language Model Explanations*. ACL 2026. [10.48550/arXiv.2509.25844](https://arxiv.org/abs/10.48550/arXiv.2509.25844)
[VLM metrics design; calibration; human-subject studies; statistical analyses]
2. **Keyu He***, Qianou Ma*, Valerie Chen, Wayne Chi, Tongshuang Wu. *RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions*. ACL Demo 2026. [10.48550/arXiv.2605.01104](https://arxiv.org/abs/10.48550/arXiv.2605.01104)
[VS Code extension; full-stack TypeScript development; cloud data pipeline design; analyses of human-AI interactions]
3. Brihi Joshi*, **Keyu He***, Sahana Ramnath, Sadra Sabouri, Kaitlyn Zhou, Souti Chattopadhyay, Swabha Swayamdipta, Xiang Ren. *ELI-Why: Evaluating the Pedagogical Utility of LLM Explanations*. Findings of ACL 2025. [10.48550/arXiv.2506.14200](https://arxiv.org/abs/10.48550/arXiv.2506.14200)
[Benchmark/Dataset design; human-subject studies; web-app implementation; multi-metric evaluation of LLM explanations]
4. Huihan Li*, Arnav Goel*, **Keyu He**, and Xiang Ren. *Attributing Culture-Conditioned Generations to Pretraining Corpora*. ICLR 2025. [10.48550/arXiv.2412.20760](https://arxiv.org/abs/10.48550/arXiv.2412.20760).
[large-scale corpus analysis; Python data engineering and visualization; survey server infrastructure design]

Experience

Adobe, San Jose, CA

May 2026 – Aug. 2026 (incoming)

Machine Learning Engineer Intern

CMind, Remote

May 2025 – Aug. 2025

Data Engineer Intern

- Built a LangChain-based RAG prototype over an internal analytics DB (chunking, embeddings, FAISS indexing/retrieval, LLM answer synthesis) with metadata filters and notebook/API utilities.
- Designed CEO-speech trait metrics and an auto-evaluation pipeline with custom rubrics, achieving Fleiss' κ of 0.89–0.91 against human annotations.

University of Southern California, Los Angeles, CA

Sep. 2022 – May 2025

Teaching & Grading Assistant

- Coordinated logistics and grading for two CS courses (CSCI-102, CSCI-360), serving ~300 students per term.
- Led weekly OHs/discussions for ~20 students, clarifying concepts and providing feedback on assignments.

Projects

AI-Based Career Advisor — University of Southern California

Nov. 2024 – Dec. 2024

- Built an AI career advisor matching user skills/interests to roles via sentence-embedding similarity over 1.3M+ job-skill entries (JobSkills, LinkedIn Jobs).
- Integrated Bing AI to surface resources and job application links, plus consistency checks on recommendations.
- Evaluated recommendations with a T5-based entailment verification model to filter low-relevance matches.

LLM Prompt Recovery Project — University of Southern California

Mar. 2024 – Apr. 2024

- Built a system to reconstruct user prompts from original text and Gemma-modified outputs.
- Fine-tuned Mixtral-8x7B with custom similarity metrics (sentence-T5-base + sharpened cosine), achieving a 0.65 leaderboard score.
- Earned a Kaggle silver medal (ranked 75/2175, **top 3.4%**) and released the final model on Kaggle.